Data Analyst Capstone Project Report

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EXECUTIVE SUMMARY

- Analysis of Stack Overflow dataset to identify key technology trends
- JavaScript and SQL dominate current and future programming usage
- PostgreSQL emerges as the leading database technology
- Cloud platforms (AWS, Azure) show strong adoption
- Developers show consistent preference trends across current and future tools
- Insights support data-driven decisions for skill development and hiring

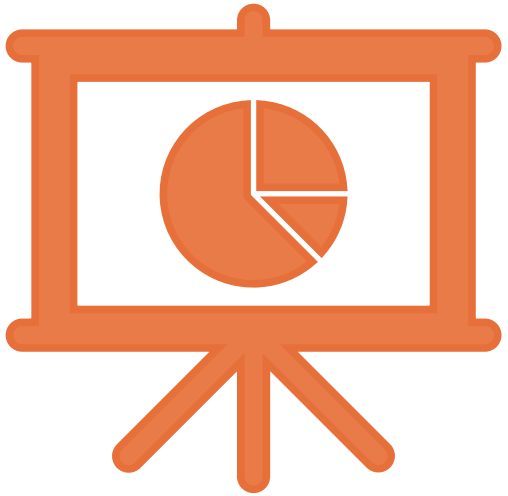
INTRODUCTION

- Objective: Analyze developer trends using Stack Overflow data
- I dentify most popular and emerging technologies
- Target audience: Data analysts, recruiters, tech leaders, learners
- Provide actionable insights for workforce and skill planning
- Highlight current vs future technology preferences
- Support strategic decision-making in tech adoption

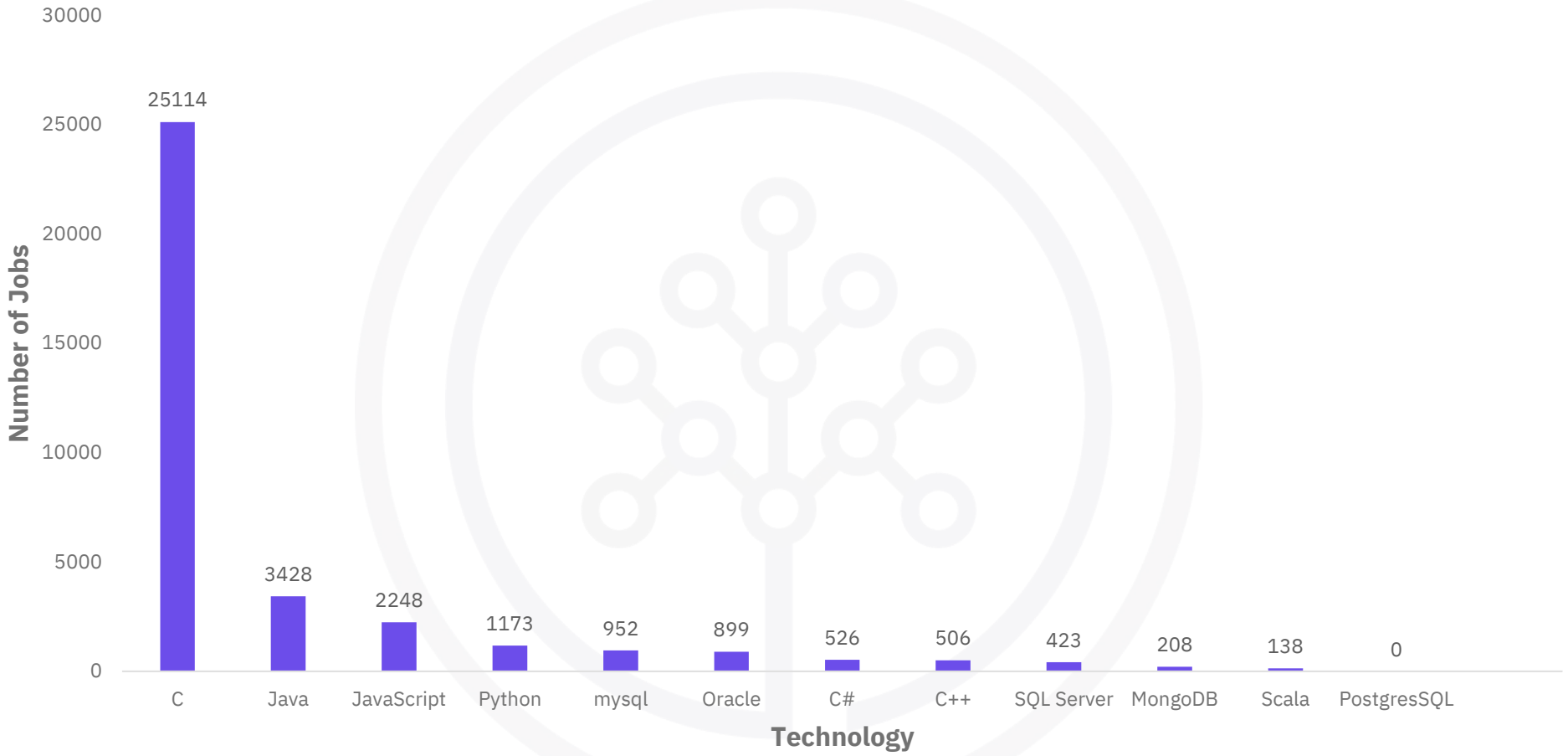
METHODOLOGY

- Data source: Stack Overflow Developer Survey dataset
- Data collection: Extracted relevant fields (languages, databases, platforms, demographics)
- Data cleaning: Handled missing values, removed duplicates, standardized formats
- Data transformation: Aggregated and grouped data for trend analysis
- Applied exploratory data analysis (EDA) techniques
- Created visualizations as Dashboards

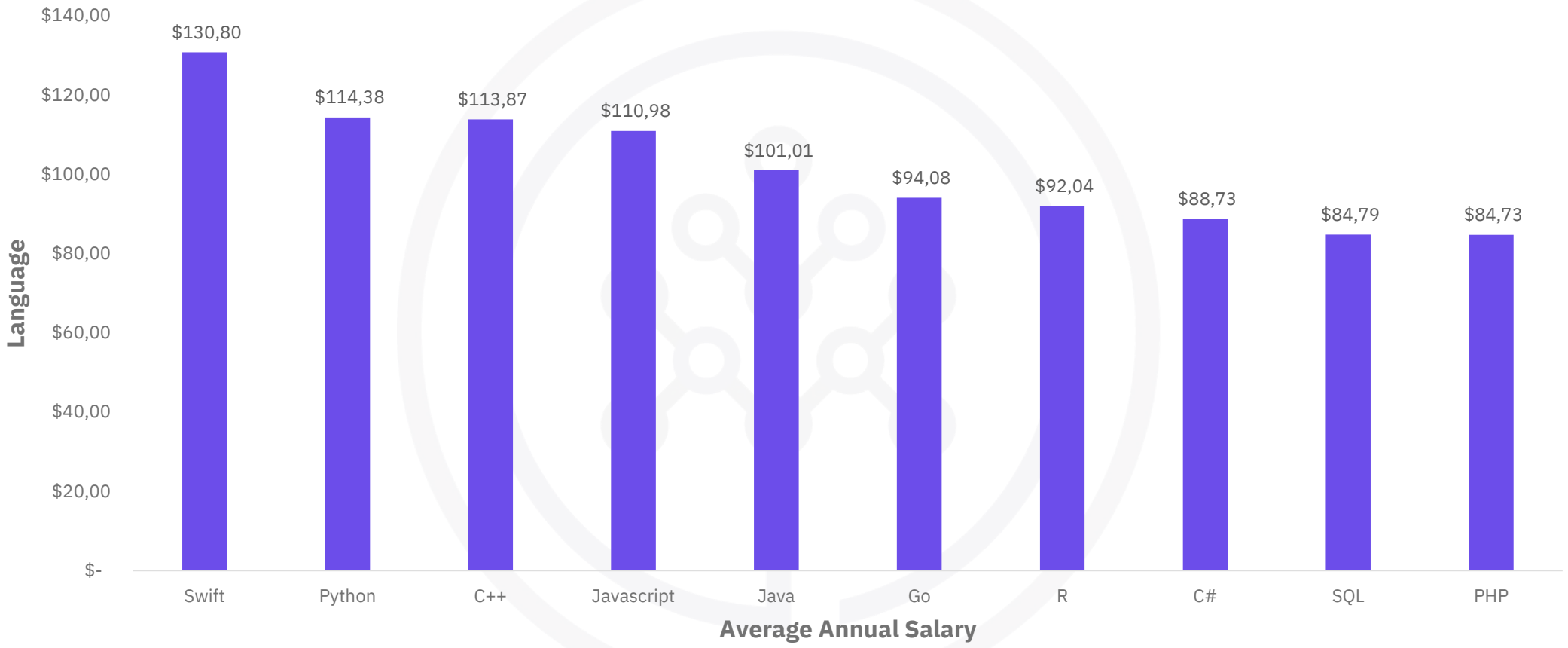
Results



JOB POSTINGS



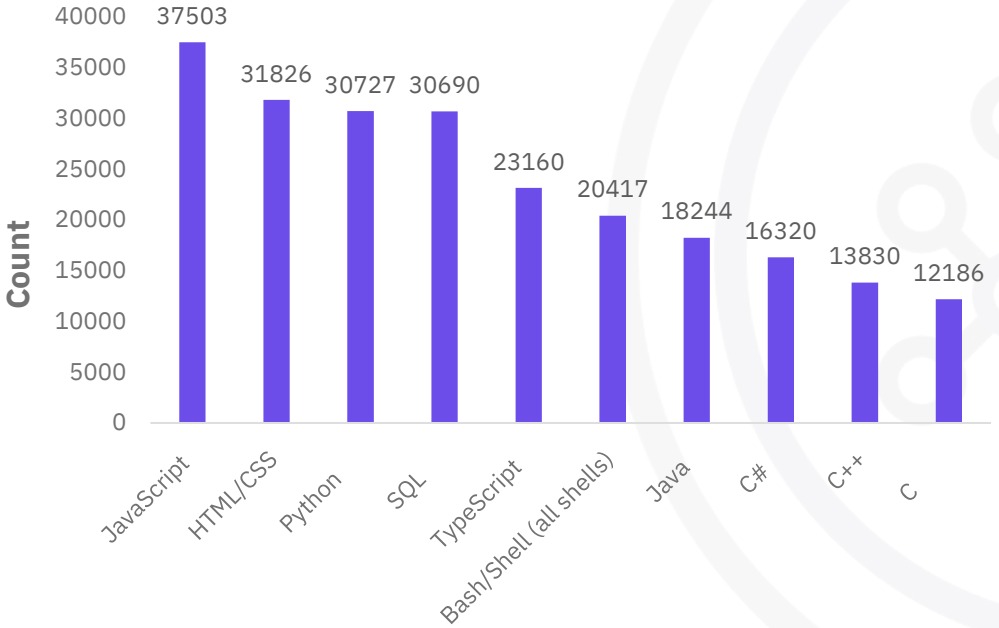
POPULAR LANGUAGES



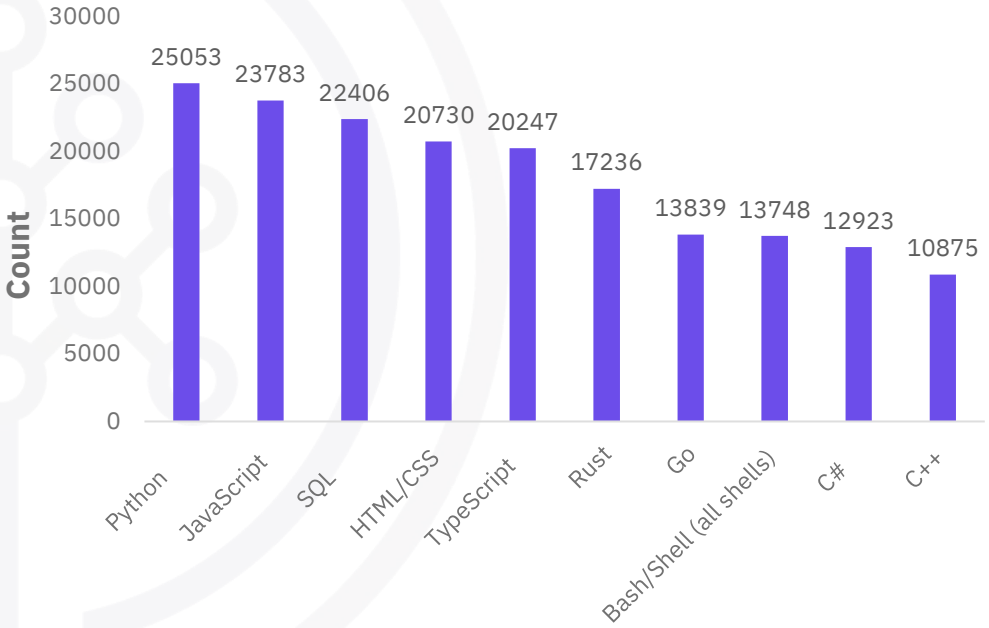
PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript remains the most widely used programming language
- SQL continues to be highly distributed due to data-driven roles
- Strong consistency between current usage and future preference

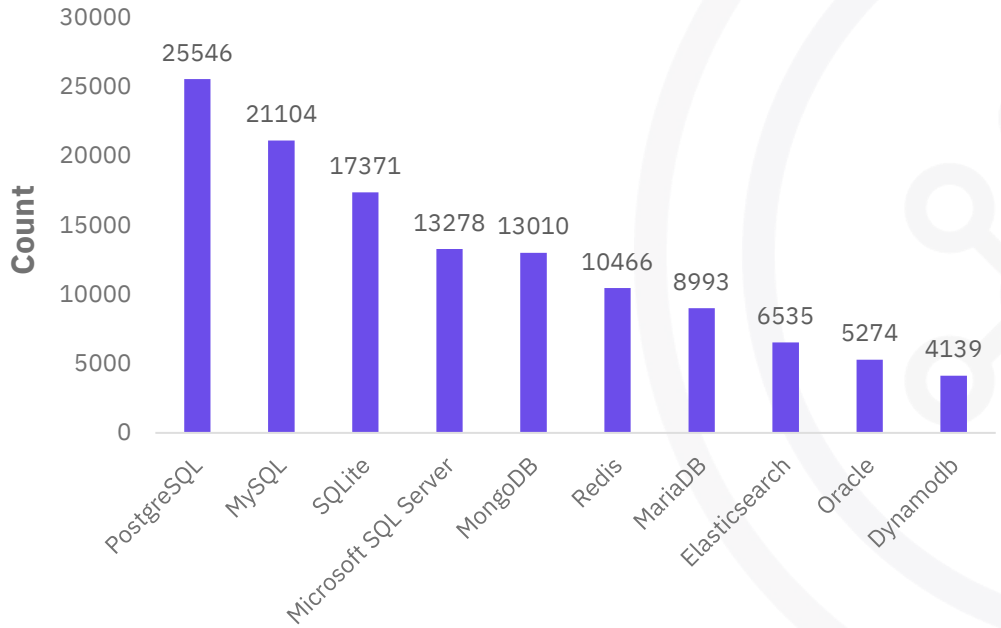
Implications

- Continued demand for JavaScript and SQL skills in the job market
- Learning these languages provides strong career stability
- Organizations should prioritize these technologies in development stacks

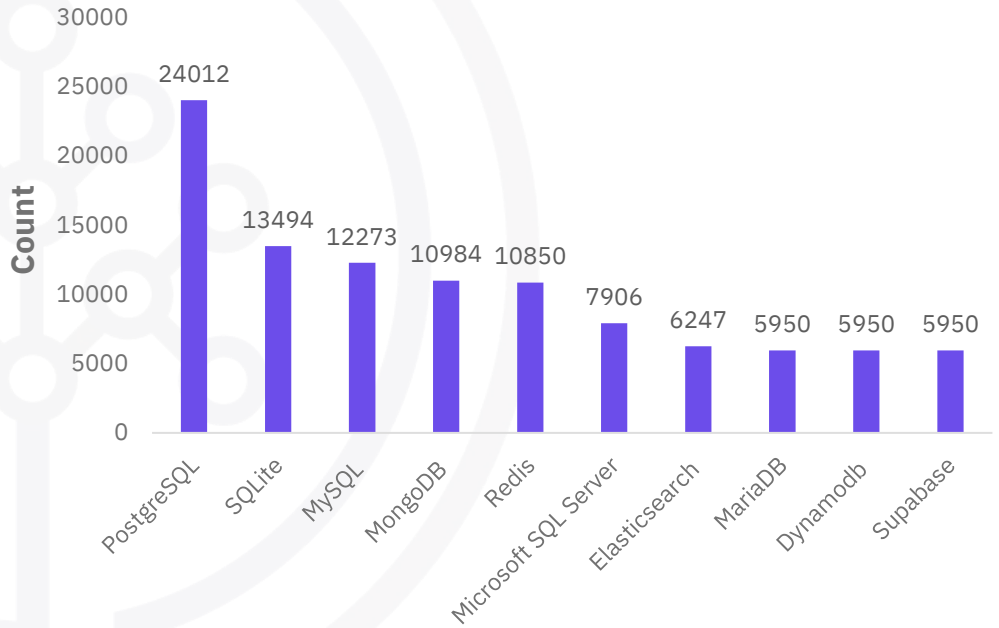
DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings:

- PostgreSQL is the most popular and fastest-growing database
- MySQL remains widely used but slightly behind
- Redis shows strong growth in future preferences

Implications:

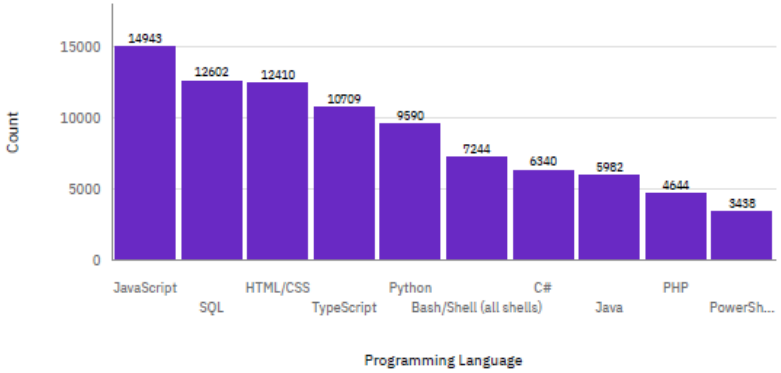
- Increasing demand for PostgreSQL expertise
- Shift toward modern, scalable database solutions
- Organizations should consider adopting high-performance databases like PostgreSQL and Redis

DASHBOARD

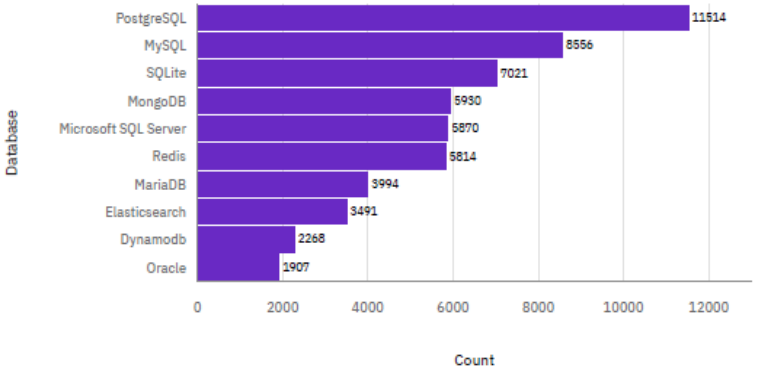
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

Current - Top 10 Programming Languages

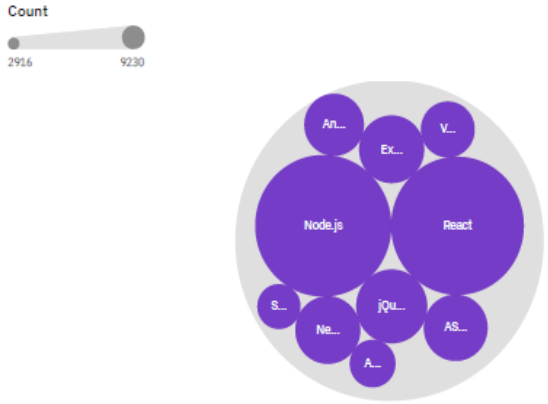


Current - Top 10 Databases



Current - Top 10 Plattforms

Current - Top 10 Webframes



DASHBOARD TAB 1: Current Technology Usage

Insights:

1. Programming Language JavaScript has the highest Count at nearly 15 thousand, followed by SQL at nearly 13 thousand
2. Database PostgreSQL has the highest Count at nearly 12 thousand, followed by MySQL at over 8500
3. Top 10 PlatformHaveWorkedWith Amazon Web Services (AWS) has the highest Count at nearly 11 thousand, followed by Microsoft Azure at over 6500.
4. Top 10 WebFrameHaveWorkedWith Node.js has the highest Count at over nine thousand, followed by React at nearly nine thousand.

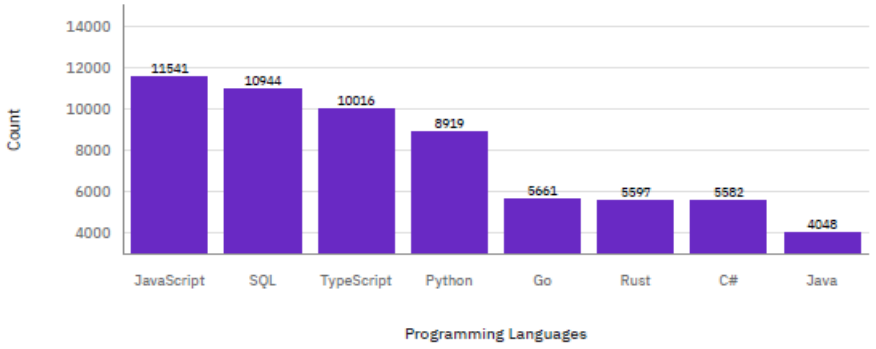
Summary Insights:

- JavaScript and SQL dominate current programming usage
- PostgreSQL leads among databases, followed by MySQL
- AWS and Azure are the most widely used cloud platforms
- Node.js and React are the top web frameworks
- Indicates strong alignment with modern web and cloud ecosystems

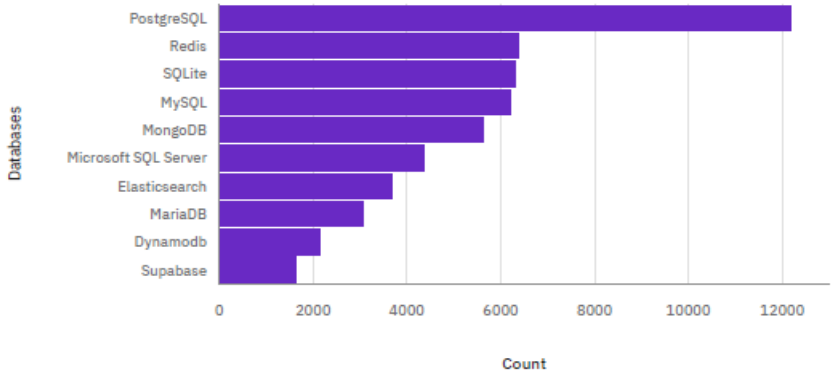
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

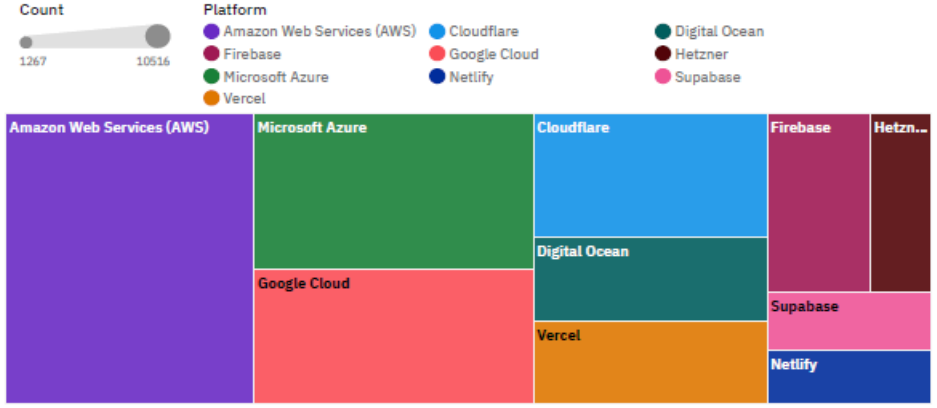
Predicted - Top 10 Programming Languages



Predicted - Top 10 Databases



Predicted - Top 10 Platforms



Predicted - Top 10 Webframes

DASHBOARD TAB 2: Future Technology Trends

Insights:

1. Programming Languages JavaScript has the highest Count at nearly 12 thousand, followed by SQL at nearly 11 thousand.
2. Databases PostgreSQL has the highest Count at over 12 thousand, followed by Redis at nearly 6500.
3. Platform Amazon Web Services (AWS) has the highest Count at nearly 11 thousand, followed by Microsoft Azure at nearly 6500
4. op 10 WebframeWantToWorkWith React has the highest Count at 8223, followed by Node.js at 8096

Summary Insights:

•JavaScript and SQL expected to remain dominant

• PostgreSQL strengthens its leading position

• Redis gains popularity as a future-focused database

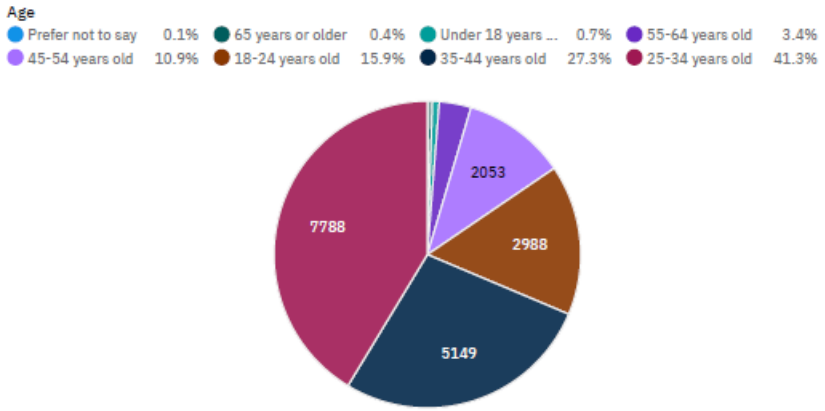
• AWS and Azure continue to lead cloud adoption

• React slightly surpasses Node.js in future preference

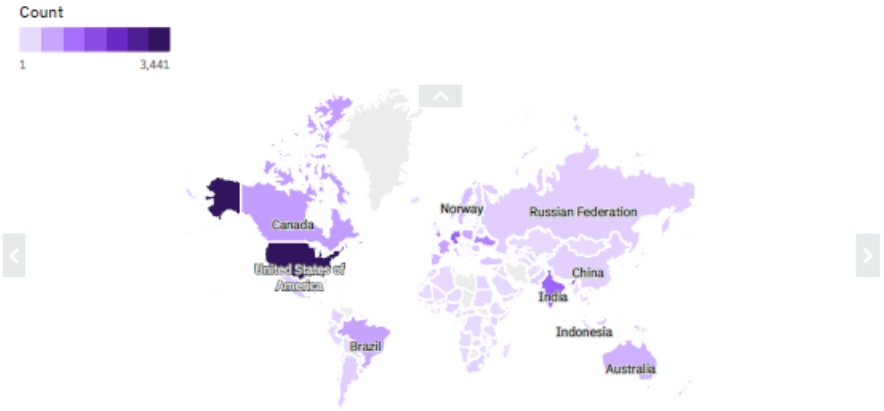
DASHBOARD TAB 3: Demographics

Demographics

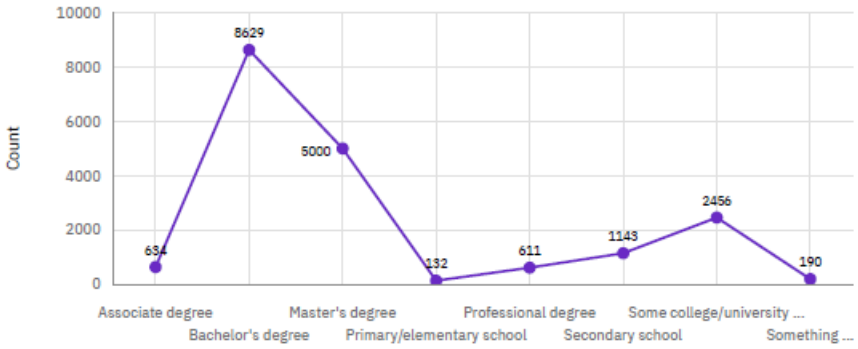
Respondent distribution by Age



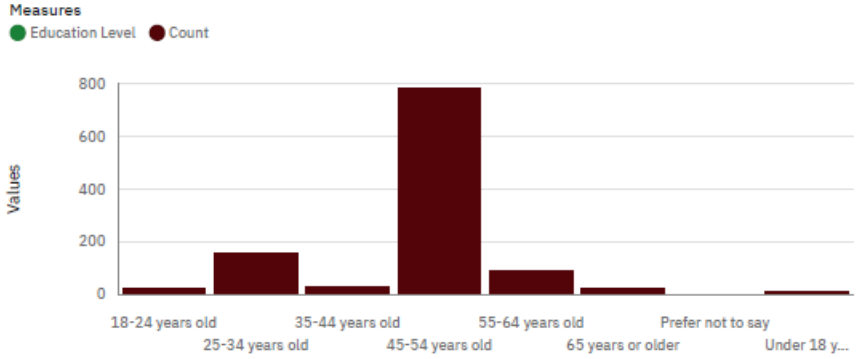
Respondent Count by Country



Respondent distribution by Formal Education Level



Respondent Count by Age, classified by Education Level



DASHBOARD TAB 3: Demographics

Insights:

1. Age 25-34 years old has the highest procentage of users.
2. America has the highest procentage of users
3. Bachelor students are mostly represented

Summary Insights:

•Majority of developers are aged 25–34

•Highest representation comes from the Americas

•Bachelor’s degree is the most common education level

•Reflects a young, educated, and globally distributed workforce

DISCUSSION

OVERALL FINDINGS & IMPLICATIONS

Findings:

- Consistent dominance of JavaScript, SQL, and PostgreSQL
- Cloud platforms are central to modern development
- Developer preferences remain stable across time

Implications:

- Strong need for cloud and full-stack development skills
- Training programs should focus on in-demand technologies
- Businesses should align tech stacks with industry trends

CONCLUSION

- Stack Overflow data reveals clear and stable technology trends
- JavaScript, SQL, and PostgreSQL are key technologies to prioritize
- Cloud computing continues to shape the future of development
- Insights provide guidance for career growth and business strategy

APPENDIX